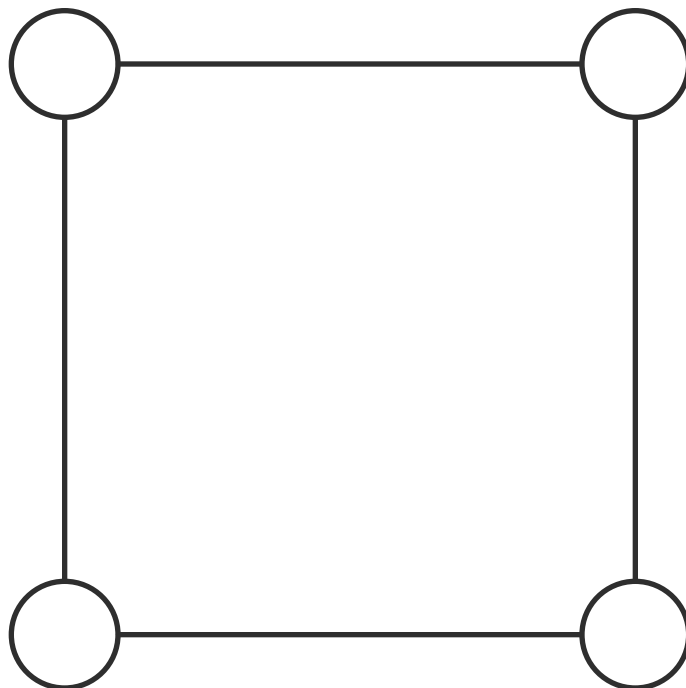
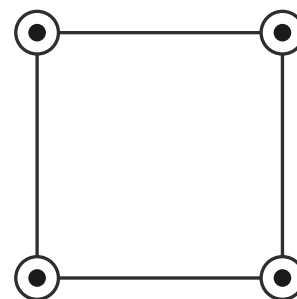
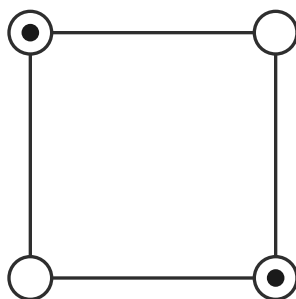
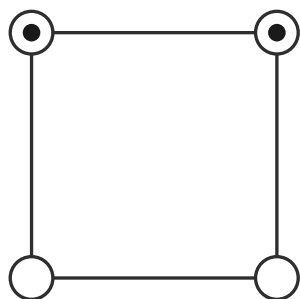


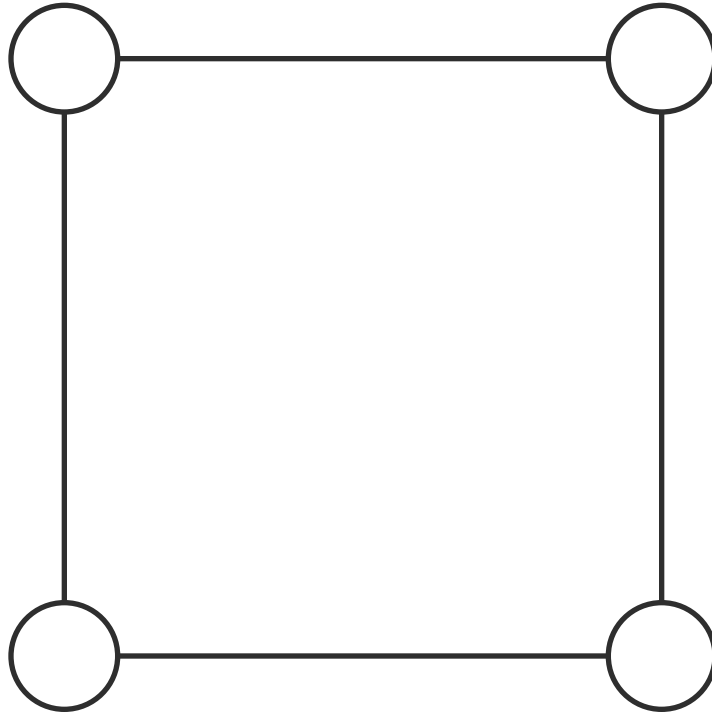
Problem 1: Empty is OFF. A dot is ON. Choose a line. Change BOTH lamps at its ends: add a dot or erase a dot. Try a move. Try that same move again.



Problem 2: Start all OFF each time. Make each small picture on the big board.

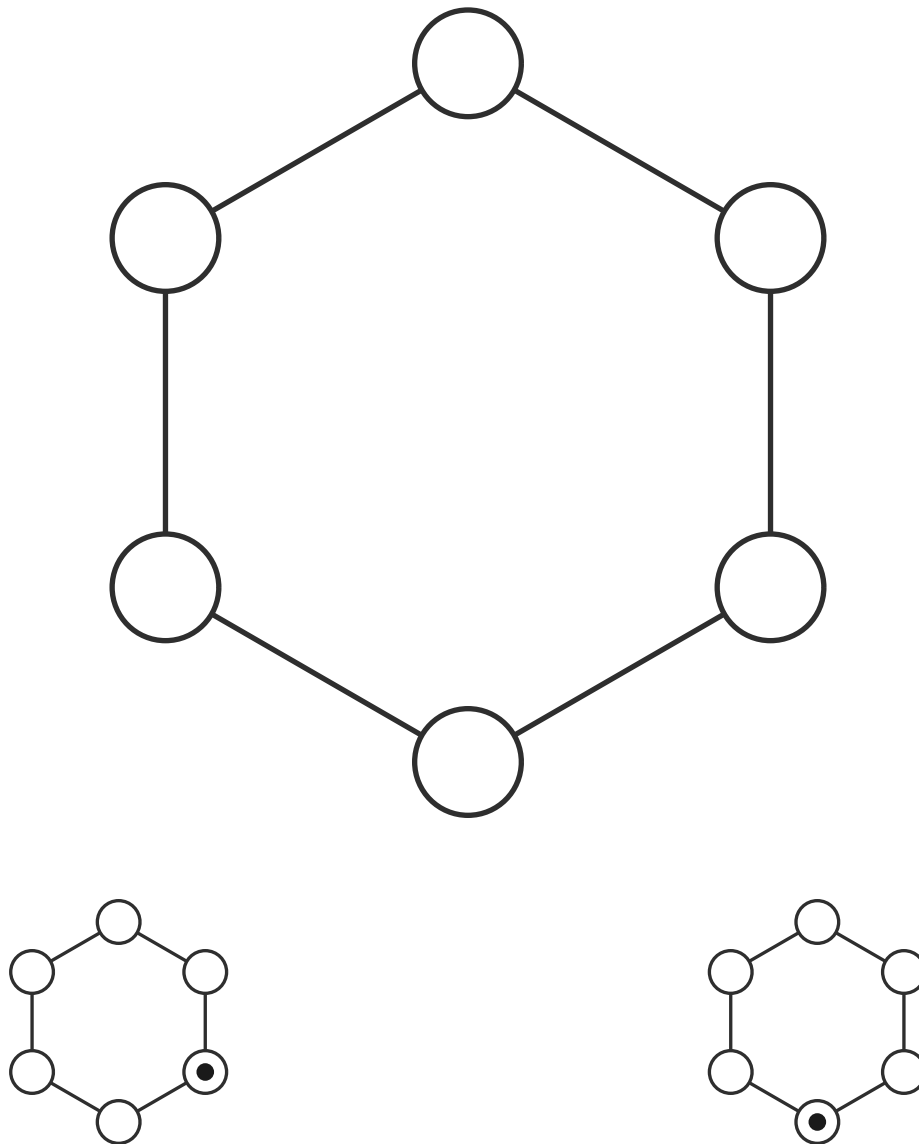


Problem 3: Play with a partner. Start all OFF. Make one move: change both ends of a line. Can your partner turn all the lamps OFF? Swap jobs.



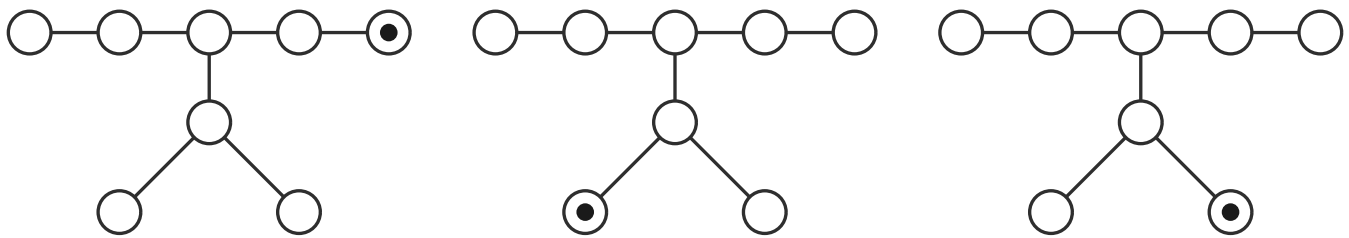
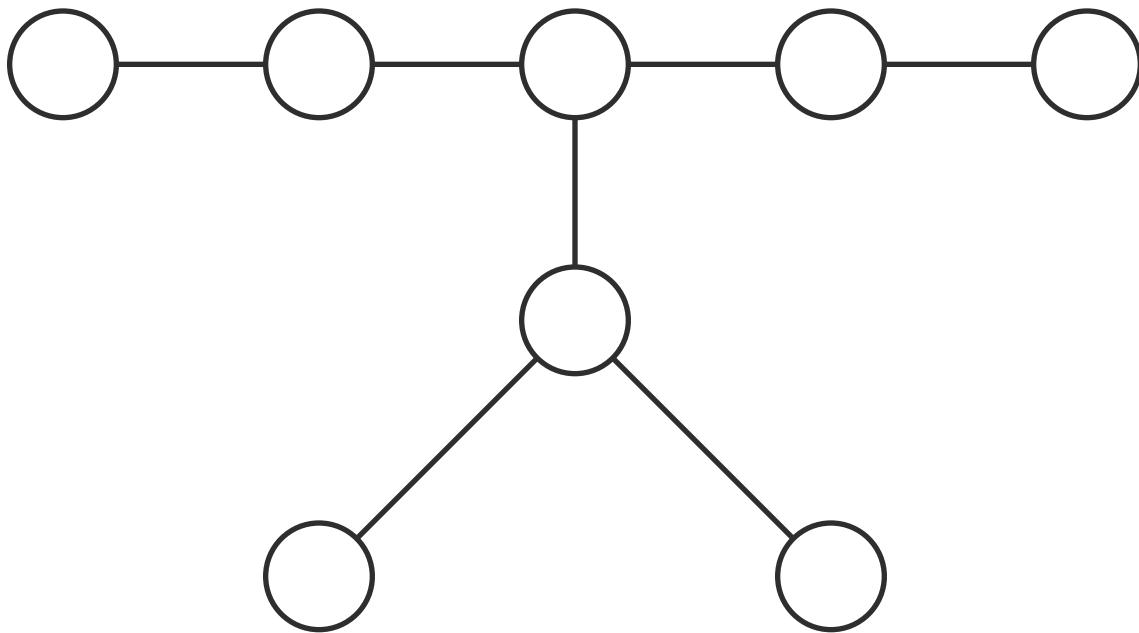
Problem 4: Start all OFF. Make TWO moves for your partner to undo. Swap jobs. Then try THREE moves. Show your moves if your partner wants a hint.

Problem 5: Start with only the top lamp ON each time. Change both ends of a line to make each small picture. Keep exactly ONE lamp ON after every move.

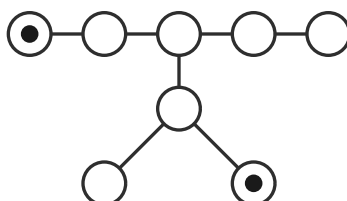


Problem 6: Start at the top again. Visit every lamp and bring the light home.

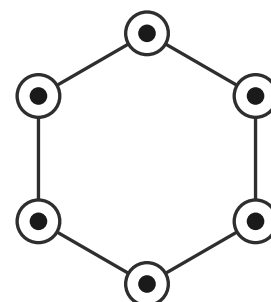
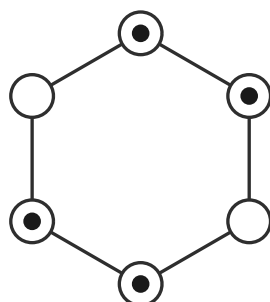
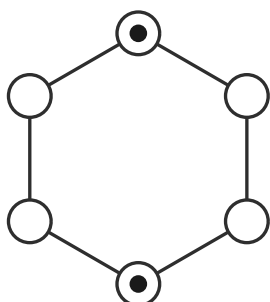
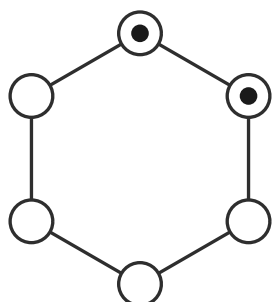
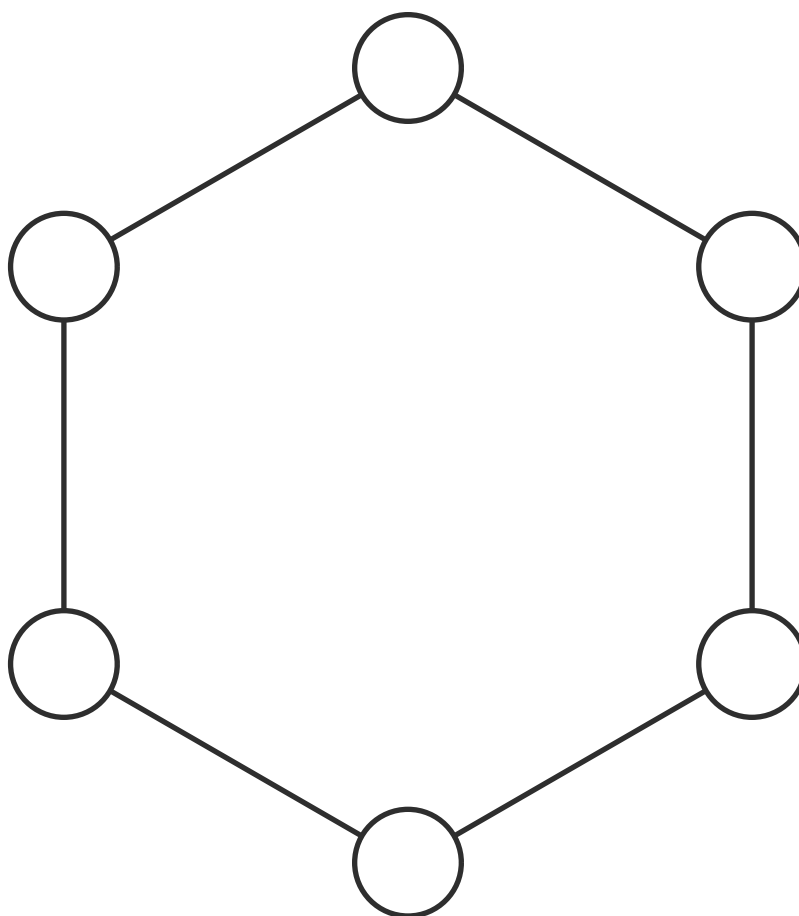
Problem 7: Start with only the far-left lamp ON each time. Change both ends of a line. Move the light to match each small picture. Keep ONE lamp ON.



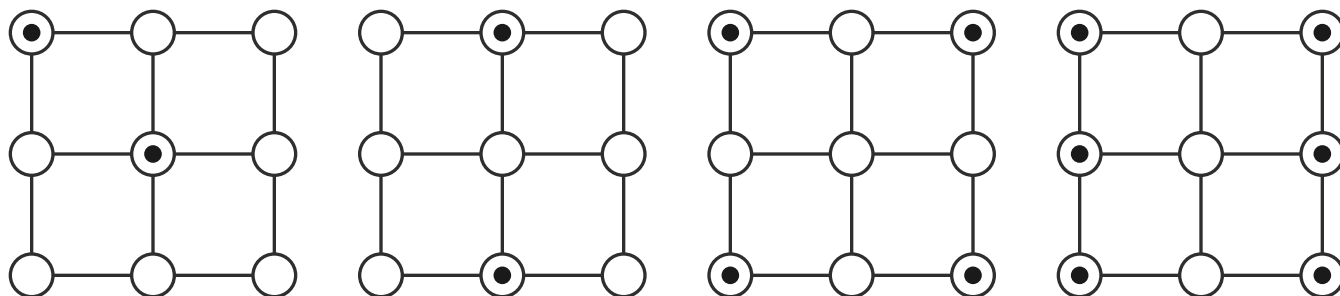
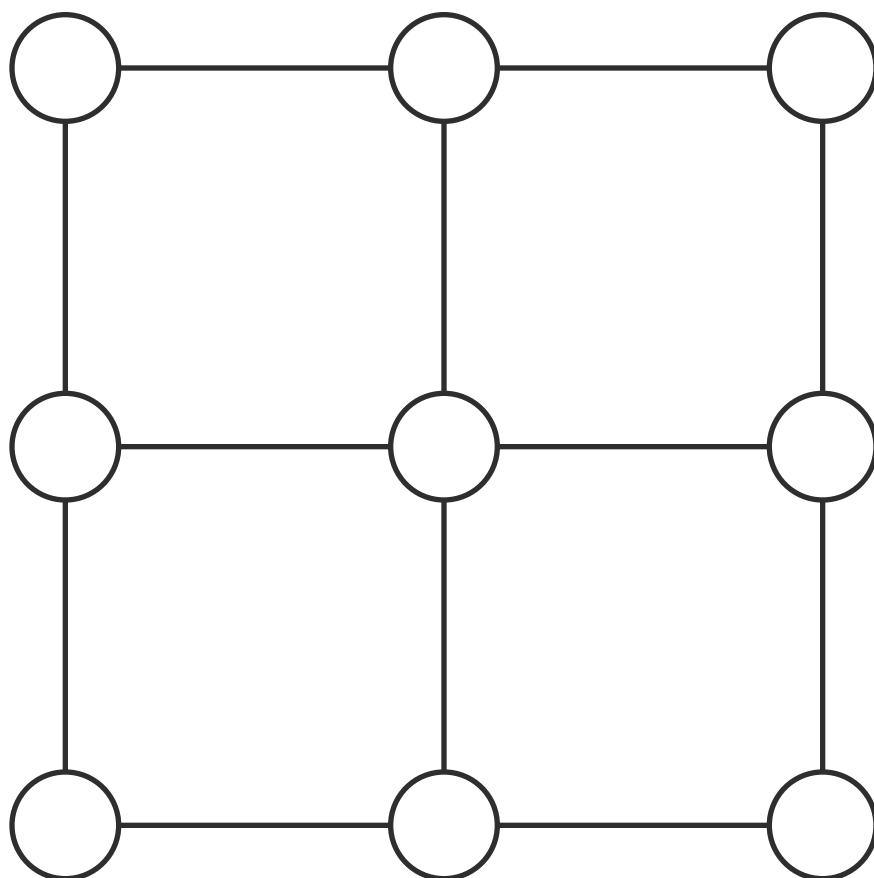
Problem 8: Now start all OFF. Make this two-light picture. You may have more than one light ON.



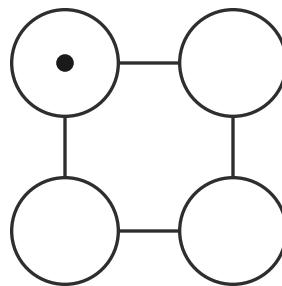
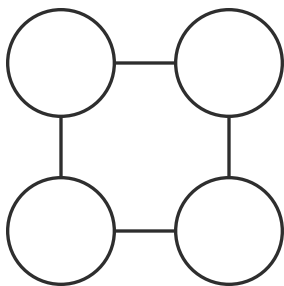
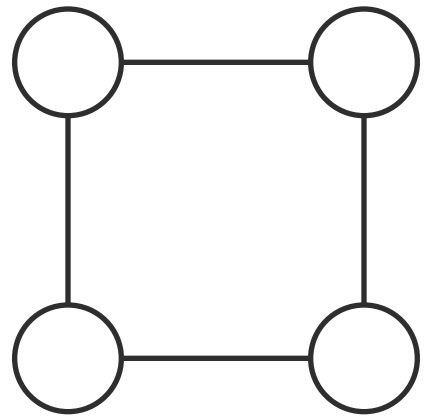
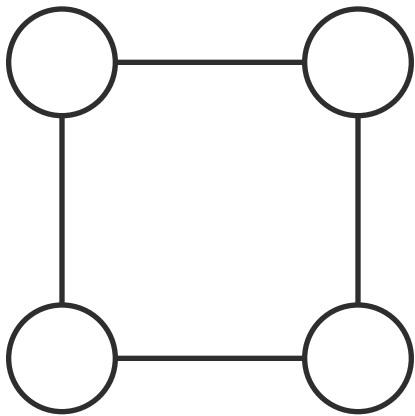
Problem 9: Start all OFF for each small picture. Change BOTH ends of one line each move. Make the picture on the big board. Point to one you want to do again.



Problem 10: Start all OFF for each small picture. Change BOTH ends of one line each move. Make the picture on the big board. Try a different way to make one picture.



Problem 11: Put a dot in the top-left lamp of the LEFT island. Change BOTH ends of one line each move. Can you make the small picture?



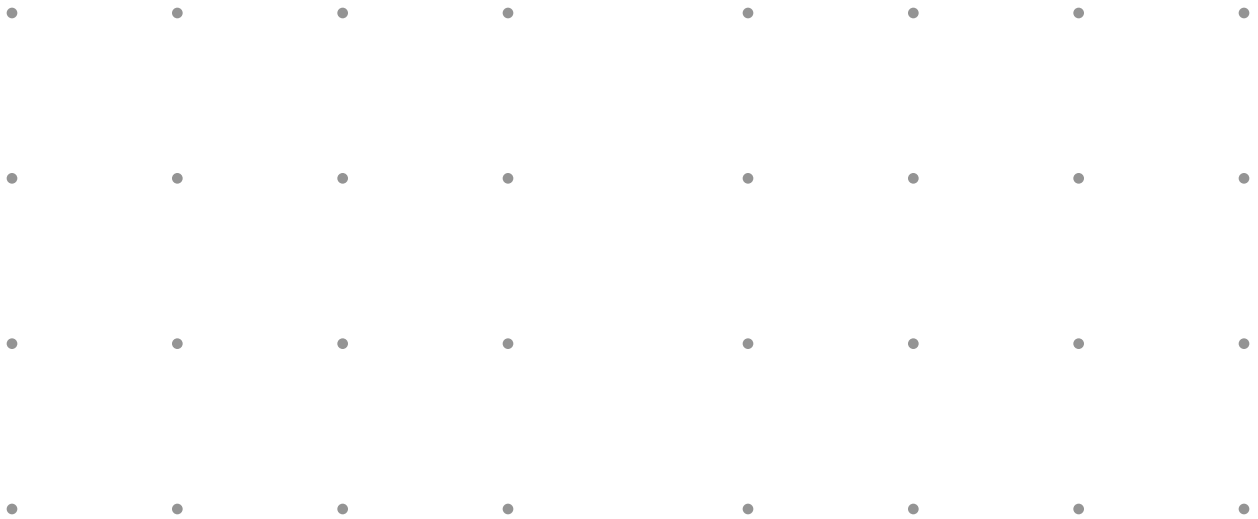
Problem 12: Draw ONE new line between lamps on different islands. Start again with only the top-left lamp of the left island ON. Try the trip with your bridge.

Problem 13: Draw 4 to 8 lamps. Join them with lines so every lamp has a path to every other lamp. Keep lines from crossing. Start all OFF. Make TWO moves, changing both ends of a line each time. Let a partner turn them all OFF.

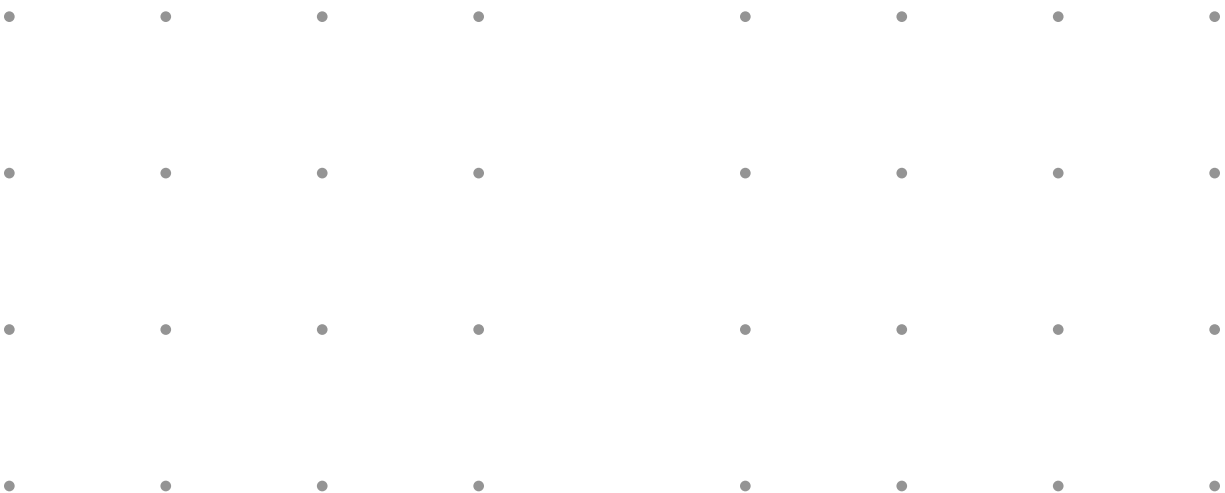


Problem 14: Add one new lamp. Join it to an old lamp with a line. Make a new puzzle for your partner.

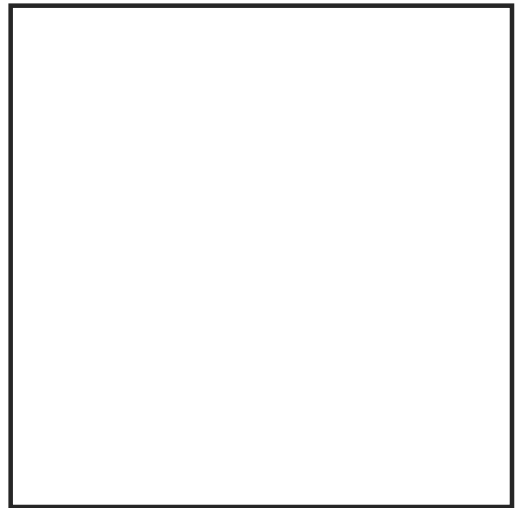
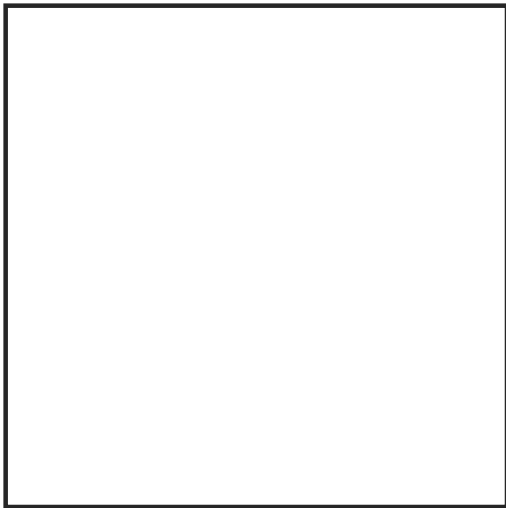
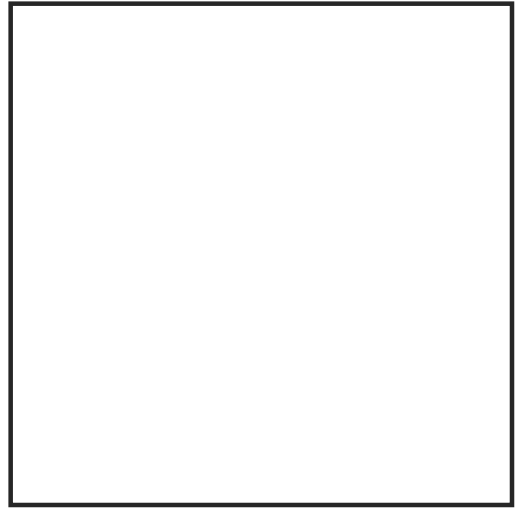
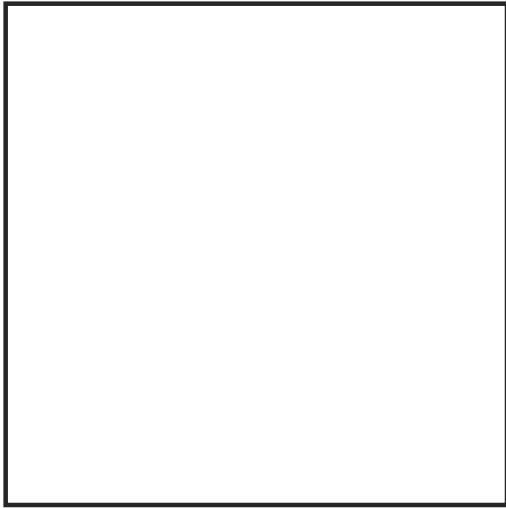
Problem 15: Join dots with straight lines to make a closed shape with THREE corners. Keep your lines from crossing. Make two different shapes.



Problem 16: Now make two different closed shapes with FOUR corners. Touch each corner where your line turns.

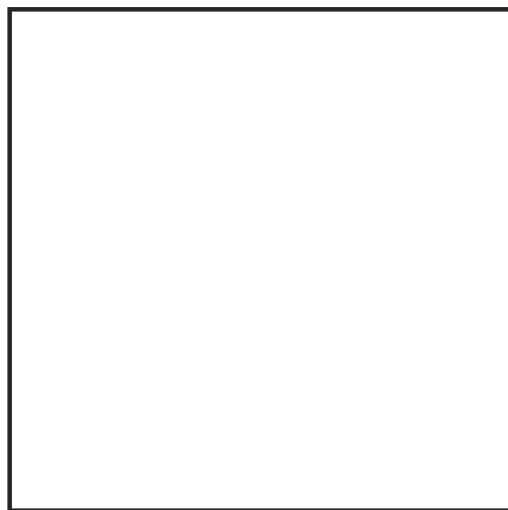
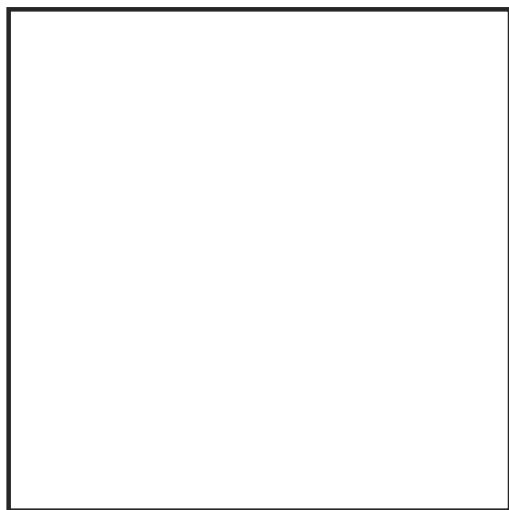
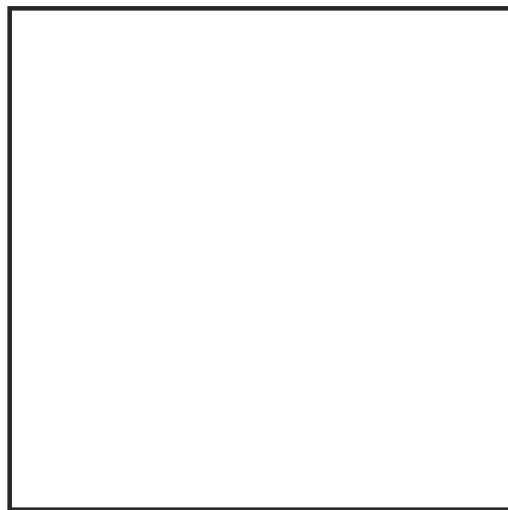
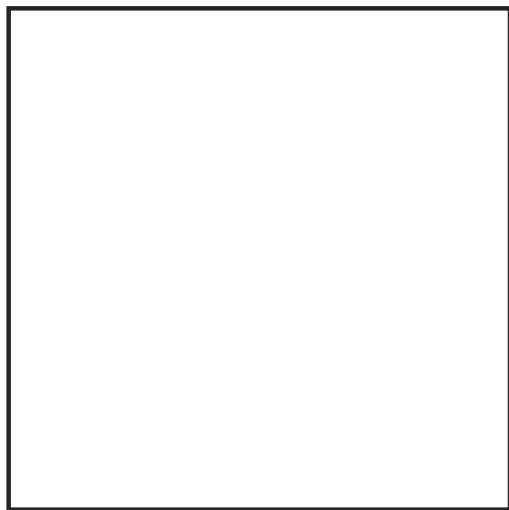


Problem 17: Each square is a house. Draw ONE straight wall all the way across each house, from its outside edge to its outside edge. Make two rooms. Try different walls.

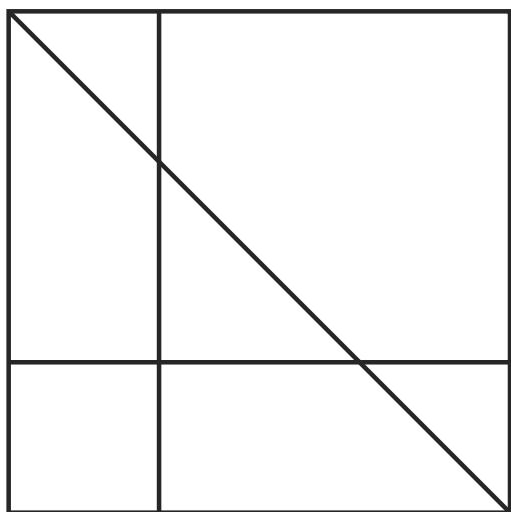
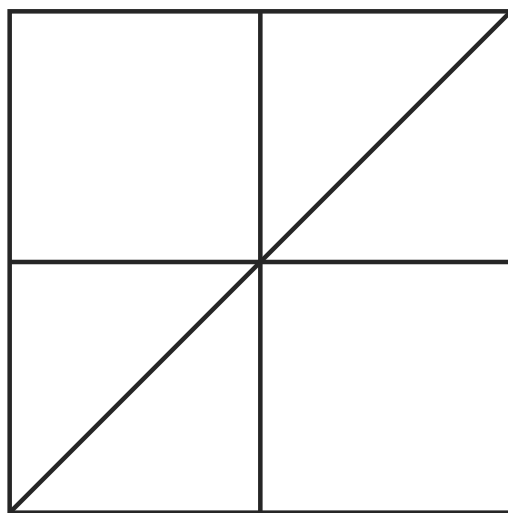
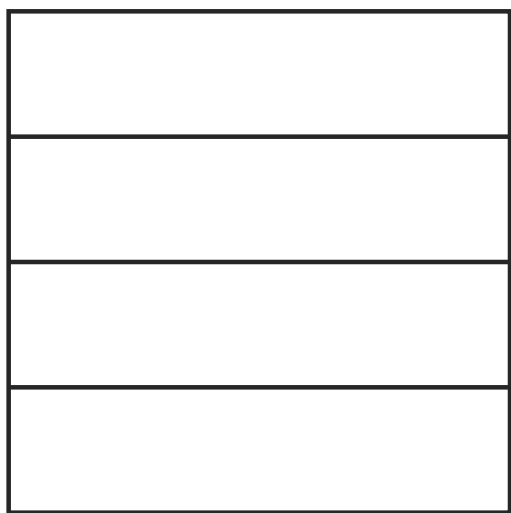


Problem 18: Make or find a house with two triangle rooms. Make or find one with two four-corner rooms. Draw on another sheet if you want to try more houses.

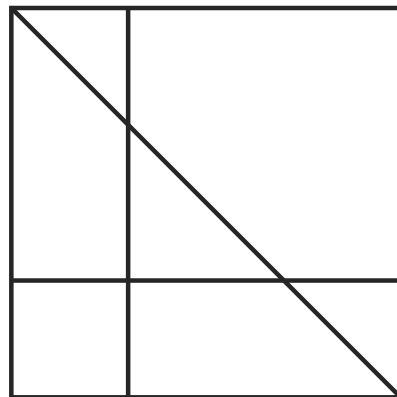
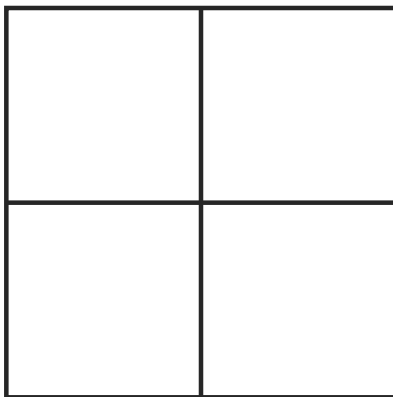
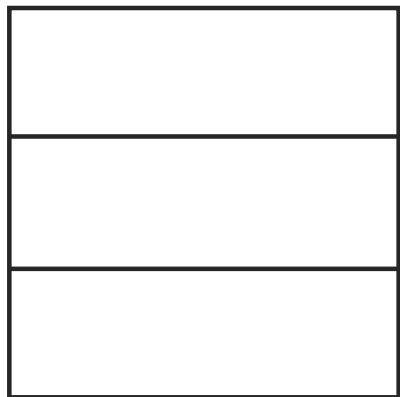
Problem 19: Draw TWO different straight walls all the way across each house. Walls may cross. Put one dot in each room. Try to make different numbers of rooms.



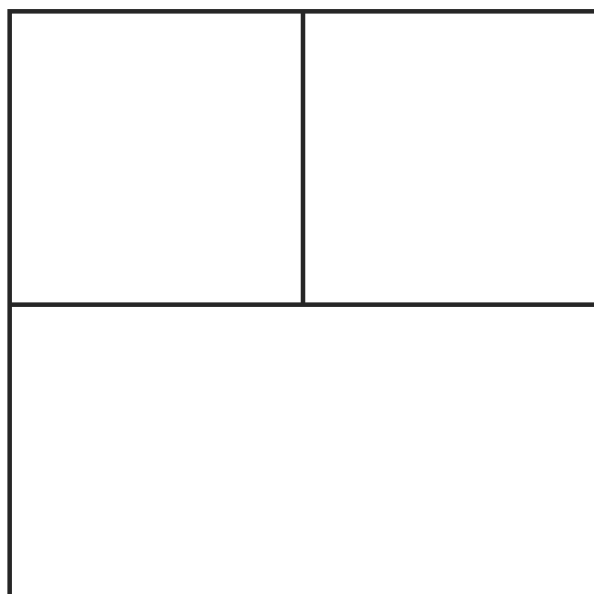
Problem 20: These houses each have THREE straight walls. Put one dot in every room. Which has the most rooms? In the empty house, try three different walls of your own.



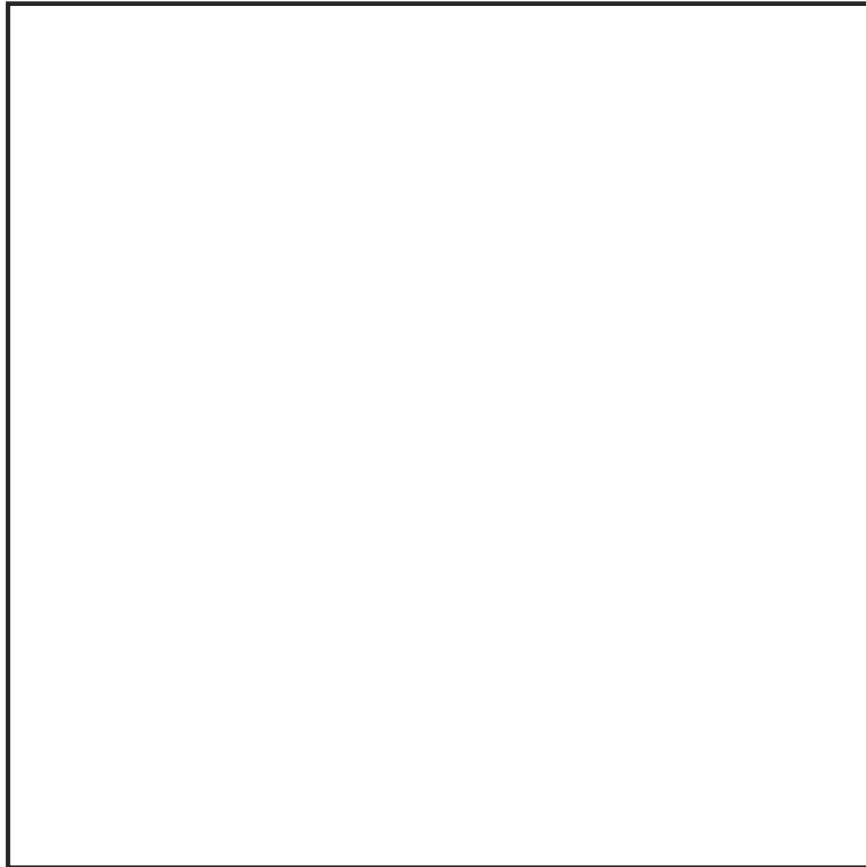
Problem 21: Put a dot or an X in every room in the first three houses. Rooms sharing a wall must have DIFFERENT marks. Rooms touching only at a corner may match.



Problem 22: Try this house. If two marks do not work, you may also use an O.



Problem 23: Make a house with one, two, or three different straight walls. Draw each wall all the way across. Let a partner put one dot in every room. Swap jobs.



Problem 24: Make a giant house together on a whiteboard or another sheet. Take turns adding one new straight wall all the way across. Do not trace an old wall. Stop after three walls. Find every room together.